

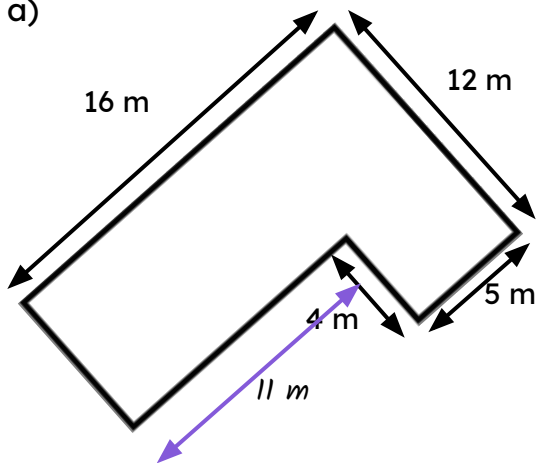


Calculating missing side lengths from the area of composite rectilinear shapes

Task A: Calculating area with missing side lengths

1) Calculate the area of each composite rectilinear shape.

a)

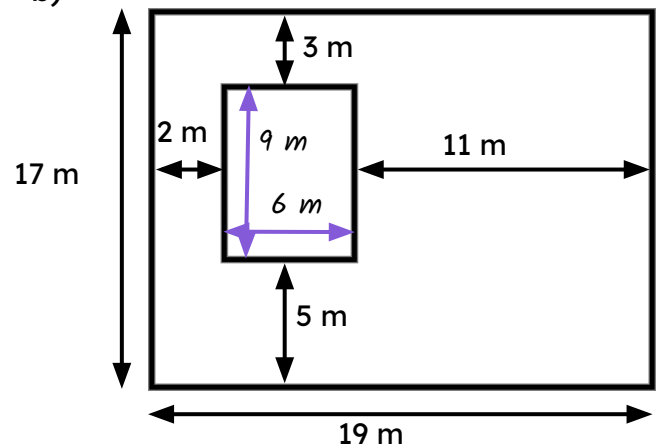


$$\text{Area} = 16 \times 12 - 11 \times 4$$

$$\text{Area} = 192 - 44$$

$$\text{Area} = 148 \text{ m}^2$$

b)



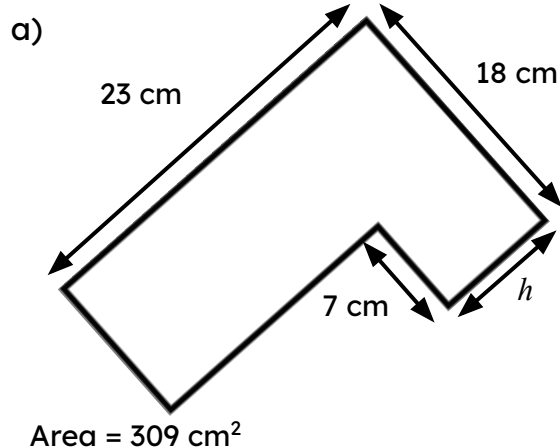
$$\text{Area} = 17 \times 19 - 9 \times 6$$

$$\text{Area} = 323 - 54$$

$$\text{Area} = 269 \text{ m}^2$$

Task B: Finding a missing side length using area

1) Calculate the length of the marked side.



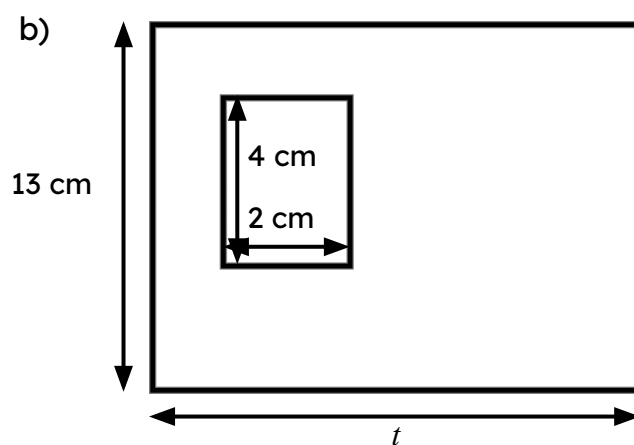
$$309 = 23 \times 11 + 7h$$

$$309 = 253 + 7h$$

$$56 = 7h$$

$$8 = h$$

The marked side is 8 cm long.



$$187 = 13t - 4 \times 2$$

$$187 = 13t - 8$$

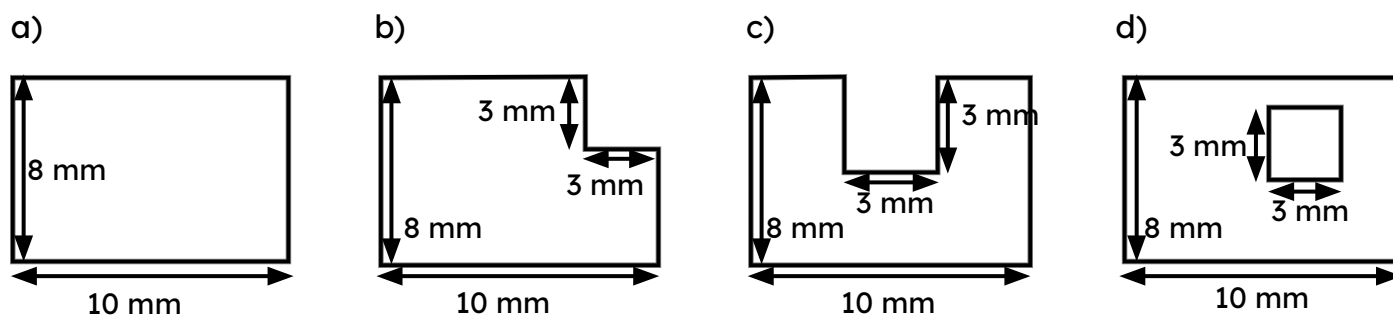
$$195 = 13t$$

$$15 = t$$

The marked side is 15 cm long.

Task C: Working with perimeter and area

1) Which shapes have the same area? Which have the same perimeter?



Shape A has an area of 80 mm^2 . Shapes B, C and D all have the same area of 71 mm^2 .

Shapes A and B both have a perimeter of 36 mm. Shape C has a perimeter of 42 mm. Shape D is ambiguous - do you want to include the square or not?